Smart Room Monitor: An RTOS-Based Environmental Sensing and Alert System

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1. Application and Use Case

The Smart Room Monitor continuously monitors a room's temperature, humidity, light level, and air quality, then responds when conditions become uncomfortable or potentially unhealthy. Instead of relying on a user to notice changes manually, the system provides live feedback and can react automatically when needed.

Why: Indoor environmental conditions can affect comfort, sleep, concentration, and the overall quality of a space. Excess humidity may encourage mold growth, while poor air quality or high temperature can make a room feel stuffy and uncomfortable. The project gives users a simple way to view these conditions in real time and receive alerts before they become more serious.

What it does:

- Displays live environmental readings on an OLED and lets the user switch screens with a button.
- Uses a PIR motion sensor to detect occupancy and enters Sleep mode when the room is empty.
- Provides RGB LED and buzzer alerts when measurements approach or exceed thresholds.
- Automatically activates a small fan when temperature or air quality becomes too high.
- Logs sensor readings to a PC through UART.

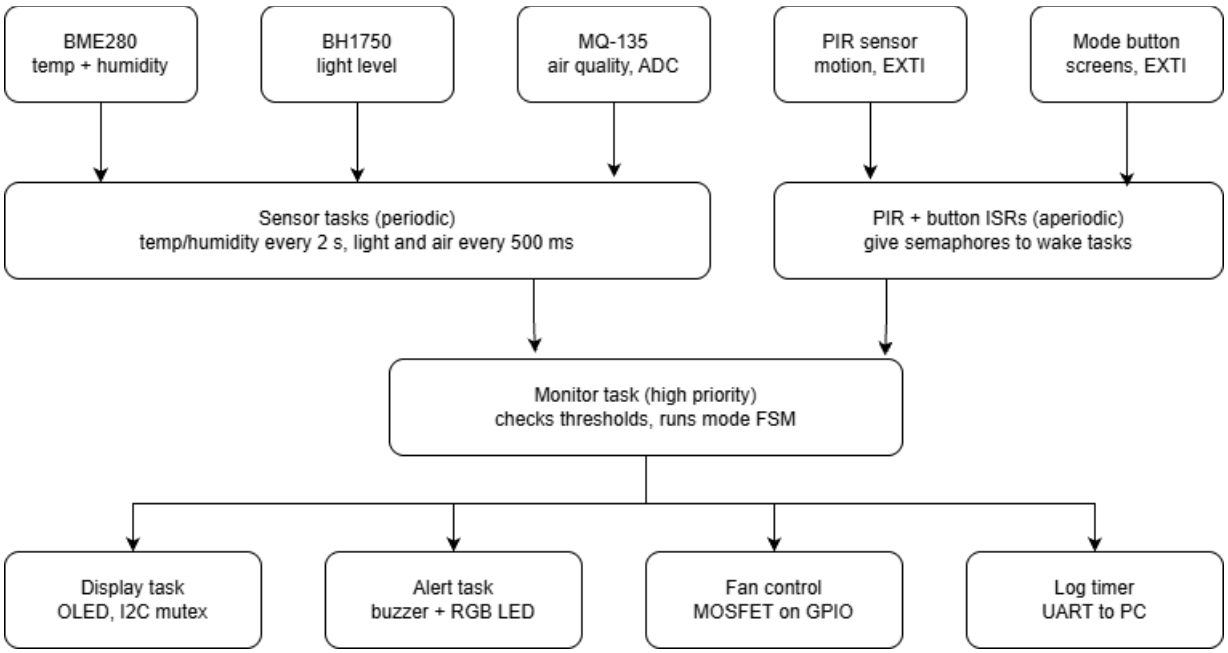
Where: Dorm rooms, bedrooms, study spaces, small server closets, and greenhouses.

Use-Case Summary

The system combines continuous environmental sensing, event-driven occupancy detection, alerts, and automatic ventilation in one embedded platform. A user can quickly check current room conditions on the OLED, while the RGB LED and buzzer provide immediate status feedback. When temperature or air quality crosses a threshold, the fan can respond automatically. If the room is unoccupied, the system reduces activity through Sleep mode to avoid unnecessary operation.

2. System Design

The system uses an STM32 Nucleo running FreeRTOS, with independent sensing, processing, and actuation tasks. Sensor tasks send readings through a queue to a high-priority monitor task that manages the FSM and system outputs.



Hardware components

Component	Role / Interface	Component	Role / Interface
STM32 Nucleo (F401RE/F446RE)	Main MCU; FreeRTOS; ST-LINK	Push button	Screens / silence alarms; GPIO / EXTI
BME280 (or DHT22)	Temperature + humidity; I2C	SSD1306 OLED	Live display; I2C
BH1750 (or photoresistor)	Light level; I2C / ADC	Piezo buzzer	Audible alerts; PWM
MQ-135	Air quality; ADC	RGB LED	Green/yellow/red status; GPIO
PIR motion sensor	Occupancy; GPIO / EXTI	DC fan + MOSFET / perfboard	Ventilation; GPIO; soldered build

Software architecture

- Embedded C in STM32CubeIDE using HAL/CubeMX.
- FSM states: Normal (green), Warning (yellow), Alarm (red + buzzer + fan), Sleep (display off, reduced sampling; motion returns to Normal).

RTOS components

- Periodic tasks at different rates (2 s and 500 ms).
- Queue sends sensor readings to monitor task.
- Priority scheduling keeps monitor/alert above display.
- PIR/button EXTI interrupts release semaphores.
- I2C mutex protects shared sensors/OLED bus.
- Software timers handle Sleep timeout and UART logging.
- Tasks block on delays, queues, or semaphores instead of polling.

Why RTOS: Sensors run at different rates, motion/button events are asynchronous, alerts need priority, and multiple devices share I2C, making an RTOS more suitable than a simple super-loop.

3. Plan, Timeline, and Work Distribution

Each checkpoint delivers a functional stage of the project: sensor/display functionality first, RTOS integration and a soldered prototype next, and a fully tested system for the final submission.

Timeline

Weeks	Dates	Goals
1	Sep 28-Oct 4	Order parts; set up STM32CubeIDE/FreeRTOS; read one sensor in a task.
2	Oct 5-Oct 11	All sensors and OLED working; submit Checkpoint 1 (Oct 11).
3-5	Oct 12-Nov 1	Implement queues, monitor FSM, alerts, PIR interrupt, and I2C mutex; solder prototype; Checkpoint 2 + TA demo.
6-8	Nov 2-Nov 22	Add fan control, Sleep mode, UART logging, and threshold tuning; test task timing and alert latency.
9-10	Nov 23-Nov 29	Final testing, report, demo preparation; final submission (Nov 29).

Work distribution

Work is divided by RTOS tasks and FSM states so both members contribute to hardware and software, while integration, soldering, testing, documentation, and state transitions are shared.

Member	RTOS tasks	FSM states
Member 1	Temperature/humidity task; air-quality task; alert task	Warning; Alarm
Member 2	Light task; PIR/button ISRs; display task; logging timer	Normal; Sleep
Both	Monitor-task integration; soldering; testing; report; logbook	Transitions between states